



E.3 Radioactive decay

Nuclear and quantum physics · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Radioactive decay is a random, spontaneous process in which an unstable nucleus emits alpha (helium nucleus, ${}^4_2\text{He}$), beta (fast electron or positron, from neutron-proton conversion), or gamma (high-energy photon) radiation to become more stable; in all decays, mass number and atomic number (and charge) are conserved. The activity of a sample, $A = \lambda N$, decreases exponentially with time, $N = N_0 e^{(-\lambda t)}$, where λ is the decay constant (probability of decay per nucleus per unit time) and N is the number of undecayed nuclei remaining; the half-life $T_{1/2} = \ln 2 / \lambda$ is the time for the activity (or the number of undecayed nuclei) to halve, independent of the sample size or history, reflecting the memoryless, statistical nature of decay.

Worked example: follow the reasoning

A radioactive sample has a half-life of 8.0 days and an initial activity of 640 Bq. Find its activity after 24 days.

24 days = $24/8.0 = 3$ half-lives.

$A = A_0 (1/2)^n = 640 \times (1/2)^3 = 640/8 = 80$ Bq.

Answer: $A = 80$ Bq.

Quick check · multiple choice

In beta-minus (β^-) decay, a nucleus emits an electron. What happens inside the nucleus to make this possible?

A A proton converts into a neutron

B A neutron converts into a proton (with emission of an electron and antineutrino)

C A neutron is ejected directly

D Two protons fuse into a neutron

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A sample of a radioactive isotope has a decay constant $\lambda = 0.0231 \text{ day}^{-1}$. [7]
- (a) Calculate the half-life of this isotope.
 - (b) The sample initially contains 5.0×10^{20} undecayed nuclei. Calculate the number of undecayed nuclei remaining after 60 days.
 - (c) Calculate the initial activity of the sample.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $T_{1/2} = \ln 2 / \lambda = 0.693 / 0.0231 = 30.0$ days.

(b) $N = N_0 e^{(-\lambda t)} = 5.0 \times 10^{20} \times e^{(-0.0231 \times 60)} = 5.0 \times 10^{20} \times e^{(-1.386)} = 5.0 \times 10^{20} \times 0.250 = 1.25 \times 10^{20}$ nuclei.

(c) $A_0 = \lambda N_0 = 0.0231 \text{ day}^{-1} \times 5.0 \times 10^{20}$. Convert λ to per second: $0.0231 / 86400 = 2.674 \times 10^{-7} \text{ s}^{-1}$. $A_0 = 2.674 \times 10^{-7} \times 5.0 \times 10^{20} = 1.34 \times 10^{14} \text{ Bq}$.

Quick check answer

B. In β^- decay, a neutron transforms into a proton, an electron, and an antineutrino; the electron and antineutrino are emitted from the nucleus, increasing the atomic number by 1 while mass number is unchanged.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.